Since the neutron will be so important in this chapter and the next, I shall digress briefly from Zwicky and his neutron stars to describe the discovery of the neutron and the relationship of neutrons to the structure of atoms.

After formulating the "new" laws of quantum mechanics in 1926 (Chapter 4), physicists spent the next five years using those quantum mechanical laws to explore the realm of the small. They unraveled the mysteries of atoms (Box 5.1) and of materials such as molecules, metals, crystals, and white-dwarf matter, which are made from atoms. Then, in 1931, physicists turned their attention inward to the cores of atoms and the atomic nuclei that reside there.

The nature of the atomic nucleus was a great mystery. Most physicists thought it was made from a handful of electrons and twice as many protons, bound together in some as yet ill-understood way. However, Ernest Rutherford in Cambridge, England, had a different hypothesis: protons and neutrons. Now, protons were already known to exist. They had been studied in physics experiments for decades, and were known to be about 2000 times heavier than electrons and to have positive electric charges. Neutrons were unknown. Rutherford had to postulate the neutron's existence in order to get the laws of quantum mechanics to explain the nucleus successfully. A successful explanation required three things: (1) Each neutron must have about the same mass as a proton but have no electric charge, (2) each nucleus must contain about the same number of neutrons as protons, and (3) all the neutrons and protons must be held together tightly in their tiny nucleus by a new type of force, neither electrical nor gravitational-a force called, naturally, the *nuclear force.* (It is now also called the *strong force.)* The neutrons and protons would protest their confinement in the nucleus by claustrophobic, erratic, high-speed motions; these motions would produce degeneracy pressure; and that pressure would counterbalance the nuclear force, holding the nucleus steady at its size of about 10^{-13} centimeter.

Box 5.1

The Internal Structures of Atoms